

Agile Project Delivery System (APDS) Framework

Agile Project Management Measurement and Decision Framework

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1. Introduction

1.1 Purpose and Scope

Scrum provides an adaptable framework tailored for fast paced and complex initiatives, organizing work into short, time boxed sprints that deliver value incrementally. This predictable rhythm allows delivery teams to respond rapidly to changing requirements while maintaining a steady output. However, without systematic tracking of core parameters, such as timeline, budget, team collaboration, stakeholder satisfaction, and scope change, project leadership often relies on subjective intuition rather than empirical data. Isolating only one or two of these areas provides an incomplete narrative of overall performance. To evaluate project health accurately, these distinct dimensions should be analyzed collectively.

A complete health check that combines these key areas helps project teams spot deeper problems before they turn into major issues. For instance, a high delivery speed combined with low stakeholder satisfaction signals an immediate misalignment in priorities. By bringing time, cost, collaboration, satisfaction, and scope changes into a single view, scrum teams can make smarter, data driven decisions. This turns Scrum from just a process into a real tool for navigating uncertainty, where compromises are clear, improvements are backed by evidence, and the team stays focused on what matters most.

The Agile Project Delivery System (APDS) is designed as a supplementary measurement and decision-making framework tailored specifically for initiatives utilizing scrum. APDS does not alter or replace standard scrum mechanics; rather, it fully respects and preserves the underlying scrum framework while introducing pragmatic data tracking and guidance for agile project environments.

While scrum explicitly defines project structures, including ceremonies, artifacts, and roles, it deliberately leaves the specific methods of measurement open ended. Consequently, project teams frequently rely on subjective feedback to evaluate initiative health. APDS is developed to bridge this gap, introducing concrete, actionable metrics that integrate directly into the scrum lifecycle without creating administrative overhead.

The APDS framework functions essentially as a data dashboard for scrum execution. It systematically tracks four core indicators: Cumulative Delivery Pace (CDP), Resource Efficiency (RE), Scope Stability (SS), Collaboration and Satisfaction (CS).

Each indicator is evaluated and categorized into distinct Green, Amber, Red performance zones. These four scores feed into a forecasting engine and a single project health score called the Project Decision. The project decision score maps directly to a clear action level, directing the Product Owner (PO) to Proceed, Watch, Review, or Escalate.

The APDS framework fits seamlessly over the existing rhythm of scrum ceremonies. During Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective, the system defines exactly what data should be calculated, analyzed, and produced. This objective layer addresses the following questions:

- How reliably is the team delivering validated value?
- What does historical trend data indicate regarding current capacity?
- Which project adjustments are supported by empirical evidence rather than subjective impressions?

By making these implicit performance variables visible via a structured dashboard, the APDS framework helps project teams identify bottlenecks early and implement targeted interventions with high confidence. This framework aligns directly with the natural order of scrum ceremonies, detailing the specific inputs required by the APDS framework and the corresponding strategic insights generated in return.

1.2 The Value APDS Brings to Scrum

The APDS provides five capabilities that enable teams to utilize the scrum framework with greater confidence:

- **Quantifiable Confidence:** Every indicator produces a numerical value between 0 and 1. This eliminates guesswork and subjective assessments, providing objective data for decision making.
- **Data-Driven Decisions:** The project score maps directly to four defined decision levels: Proceed, Watch, Review, or Escalate. This ensures all stakeholders maintain clear visibility into project health and the required next steps.
- **Early Warning System.** Amber zones flag emerging issues early, while red zones demand immediate corrective intervention. This allows the project team to address challenges before they escalate into crises.
- **Forecast Accuracy:** One forward-looking forecasts predicts the total number of sprints needed to complete the project.
- **Governance Traceability.** Every decision is backed by documented data. This establishes an auditable, repeatable, and defensible.

1.3 Who Should Use This Framework

This framework is designed for Project Managers, product Owners, Scrum Masters, or team leads who want to back up decisions with measurable data. It works best when a team already runs scrum and leadership requires numbers to back up professional judgment without interrupting the team's daily work.

2. Core Concepts

This section covers the building blocks needed to understand the framework before putting it into practice. It introduces the four indicators used to check project health, showing how these individual scores combine into a single project score, map into performance status zones, and feed into the forecasting engine.

These core concepts consist of Work Units, the Work Progression Ladder, Fibonacci Estimation, the Four Indicators, Project Score and Decision Levels, Actual Progress and Planned Progress, Zone Classification, Forecasting Engine, and MoSCoW prioritization with APDS scoring.

2.1 Work Units (WU)

A Work Unit represents an individual requirement, such as a feature, a work package, a user story, or a task. Regardless of the internal terminology used by an organization, the Agile Project Delivery System (APDS) framework tracks every Work Unit across the project lifecycle using three elements:

- An effort estimate based on the Fibonacci sequence (fixed unless scope changes occur)
- A tracking state defined by a specific Rung assignment (R1-R5)
- A weighted progress value calculated by multiplying the Fibonacci estimate by the Rung completion percentage.

2.2 Work Progression Ladder (WPL)

To track Work Units, the framework utilizes the Work Progression Ladder. This evaluation system consists of five Rungs (R1-R5), with every Work Unit continuously mapped to a single Rung. Each level carries a fixed completion percentage that feeds directly into the cumulative actual progress calculations.

The structural distribution of the ladder is defined below:

Rung	Description	Completion %
R1: Draft	Work identified and assigned.	0%
R2: Assembly	Work is in progress. Components are being built.	25%
R3: Validation	Work complete enough for review. Testing/peer review.	60%
R4: Certified	Work has passed validation. Approved by reviewer/QC.	85%
R5: Agreed	Work fully accepted by PO/customer. No further action.	100%

Teams may calibrate Rung descriptions and completion percentages to match their workflow. Rungs are weighted to reflect increasing certainty. Earlier Rungs (R1-R2) carry lower percentages because work is still in development stage. The largest jump at R3 (35 points) reflects the transition from development into validation, which represents the highest-effort phase. Later Rungs (R4-R5) carry smaller increments because the remaining work is verification and acceptance.

2.3 Fibonacci Estimation

Every Work Unit gets a Fibonacci number that represents expected effort in days: 1 = 1 day, 2 = 2 days, 3 = 3 days, 5 = 5 days, 8 = 8 days, and so on. The Product Owner and the team agree on these estimates together during sprint planning and backlog refinement. Once set, the estimate stays fixed unless the scope of the Work Unit changes.

2.4 Zone Classification & The Four Indicators

To assess project health, APDS tracks four indicators. These are calculated at the end of each sprint (prior to the next iteration during the sprint planning session) and categorized into one of three status zones:

- **Green (On Track):** No action is required.
- **Amber (Caution):** Close monitoring is advised.
- **Red (Action Required):** Immediate intervention is needed.

Every APDS indicator scores between 0% and 100% and falls into one of three zones based on threshold ranges:

Zone	Range	Color	Meaning
On track	90% - 100%	Green	No action required. Continue monitoring.
Caution	80% - 89%	Amber	Investigate. Prepare corrective action.
At risk	0% - 79%	Red	Immediate intervention required. Escalate if needed.

The table below outlines the four indicators, and how they are measured:

Indicator	Formula	Question It Answers
Cumulative Delivery Pace (CDP)	AP_cum / PP_cum	Is delivery on track?
Resource Efficiency (RE)	SB / SC	Is cost within budget?
Scope Stability (SS)	$1 - (Current - Original / \text{Max}(Current, Original))$	Is scope stable?
Collaboration & Satisfaction (CS)	$(Collaboration * 0.4) + (Satisfaction * 0.4) + (Communication * 0.2)$	Is the team healthy?

2.4.1 Planned Progress (PP)

The logic behind Planned Progress (PP_sprint) dictates that for every Work Unit committed to a given sprint, the plan is to complete 100% of that commitment. Therefore, PP_sprint value is always 1.0 (100%) for the committed scope. The formula weights each committed Work Unit by its estimated Fibonacci duration. The formula is defined as:

$$PP_sprint = \Sigma (WU \text{ Fib} * 100\%) / \Sigma (WU \text{ Fib}) = 1.0 (100\%)$$

Example: If a team commits to three Work Units for an upcoming sprint:

Login: Fibonacci 5, **Payment:** Fibonacci 8, **Dashboard:** Fibonacci 3

$$PP_sprint = (5*100\% + 8*100\% + 3*100\%) / (5+8+3) = 1.0 \text{ (100\%)}$$

Note: If scope changes occur mid-sprint, PP_sprint is recalculated to reflect the updated set of committed Work Units.

2.4.2 Actual Progress (AP)

Actual Progress (AP) represents the true completion rate of the sprint. It is calculated by taking every Work Unit in the sprint, multiplying its Fibonacci estimate by its WPL completion percentage, summing those values, and dividing by the total Fibonacci in the sprint. The result is a number between 0% and 100% that shows how much work is done. The formula is defined as:

$$AP = \Sigma (\text{WU Fib} * \text{Rung Completion \%}) / \Sigma (\text{WU Fib})$$

Note: The project team updates the WPL status of every Work Unit during the daily scrum throughout the sprint.

Example: Consider a sprint with a total of four committed Work Units representing a combined workload of 21 Fibonacci points (the sum of all planned Work Unit durations). The following table illustrates the breakdown of individual Work Unit progress within the sprint:

Work Unit	Fib	Rung	Completion %	Weighted Progress
Login feature	5	R3	60%	3.0
Payment integration	8	R4	85%	6.8
User dashboard	3	R1	0%	0.0
Admin panel	5	R2	25%	1.25

If the sum of each individual Work Unit's Fibonacci value multiplied by its current Rung completion percentage yields a total earned progress of 11.05, the Actual Progress (AP) calculation is as follows:

$$AP = 11.05/21 = 0.526 \text{ (52.6\%)}$$

This result indicates that the team achieved 52.6% of the total planned effort committed across the specified Work Units for the sprint. This percentage is relative to a baseline where all active Work Units are at Rung R5 (100% completion), which equates to the full 21 points of effort.

Cumulative Planned Progress (PP_cum) represents the percentage of total planned work expected to be complete by the current sprint. The formula is defined as:

$$PP_cum = (100\% / \text{Total Sprints}) * \text{current sprint number}$$

Cumulative Actual Progress (AP_cum) is the cumulative weighted progress through the WPL up to the current sprint.

2.4.3 Cumulative Delivery Pace (CDP)

Cumulative Delivery Pace (CDP) measures whether the team is delivering work as fast as originally planned. It compares Cumulative Actual Progress (AP_cum) against Cumulative Planned Progress (PP_cum). By looking at total progress across all sprints rather than just one single sprint, this metric smooths out short-term fluctuations. This gives stakeholders a clear, accurate view of long-term delivery trends and schedule health.

The formula is defined as:

$$\text{CDP} = \text{AP_cum} / \text{PP_cum}$$

Example: Consider a project at the conclusion of sprint 5 out of a total 10 planned sprints:

- Cumulative Planned Progress (PP_cum) = 50%
- Cumulative Actual Progress (AP_cum) = 30%

The calculation is executed as follows: $\text{CDP} = 30\% / 50\% = 60\%$ (Critical)

The table below shows the three CDP zones and the meaning of each:

Zone	Range	Meaning
Green (Healthy)	90% - 100%	Delivery is tracking safely on schedule. The team is meeting its cumulative commitments.
Amber (Behind)	80% - 89%	Delivery is experiencing minor delays. The schedule requires close monitoring and potential adjustment.
Red (Critical)	0% - 79%	Delivery is significantly behind schedule. Immediate intervention is required to address root causes.

2.4.4 Resource Efficiency (RE)

The Resource Efficiency (RE) indicator evaluates whether the team is spending resources according to the planned budget. It compares the Sprint Budget (SB) against the actual Sprint Cost (SC). The formula is defined as:

$$\text{RE} = \text{SB} / \text{SC}$$

The Sprint Budget (SB) represents the total project budget distributed across the planned number of sprints. Sprint Budget is compared against the actual Sprint Cost (SC) to determine the project's Resource Efficiency indicator.

The formula is defined as:

$$\text{SB} = \text{Total Project Budget} / \text{Total Number of Sprints}$$

Example: Total Project Budget = \$500,000, Total Number of Sprints = 10 sprints

$$\text{SB} = \$500,000 / 10 = \$50,000 \text{ per sprint.}$$

RE cannot go above 100% If sprint spent less than planned, the extra does not count. RE above 100% indicates the sprint was delivered under budget. However, this metric should be cross checked against Cumulative Delivery Pace (CDP) to understand the root cause:

- If the CDP is Green, the under-spend reflects genuine efficiency.
- If the CDP is Red, the under-spend likely reflects incomplete work.

Example: Consider a sprint with the following parameters:

- Sprint Budget (SB) = \$50,000
- Sprint Cost (SC) = \$53,000

$RE = \$50,000 / \$53,000 = 94.3\%$ (On-Target).

The table below shows the three RE zones and meaning of each range:

Zone	Range	Meaning
Green (On-Target)	90% - 100%	Resources used efficiently.
Amber (Over-Budget)	80% - 89%	Resources are not used efficiently. Investigate cost drivers.
Red (Critical)	0% - 79%	Resources consumed with little progress. Immediate intervention.

2.4.5 Scope Stability (SS)

Scope Stability (SS) is the foundational APDS indicator utilized to evaluate whether the project scope is remaining steady or experiencing frequent changes. While baseline scope modifications typically occur during backlog refinement, this metric also accounts for scope changes driven by stakeholder feedback during the Sprint Review.

The indicator monitors scope variance by comparing the original Work Unit (WU) count against the current count. Every time a Work Unit is added to or removed from the project scope, the SS metric shifts accordingly.

The value of the change is used to ensure that both scope creep (adding work) and scope reduction (removing work) are captured as deviations from the baseline plan. The formula is defined as:

$SS = 1 - (| \text{Current WU Count} - \text{Original WU Count} | / \text{Max} (\text{Original WU Count}, \text{Current WU Count}))$

Example: Consider a project with an expanding backlog due to refinement adjustments and stakeholder review feedback:

- Original Scope: 20 Work Units
- Current Scope: 24 Work Units

Calculation: $SS = 1 - (| 24 - 20 | / \text{Max} (24, 20)) = 0.833$ (83%)

A result of 83% places the project scope within the Amber (Creeping) zone.

The table below outlines the three zones for Scope Stability (SS) and meaning of each range:

Zone	Range	Meaning
Green (Stable)	90% - 100%	The scope is stable. Minor adjustments are well within manageable project tolerances.
Amber (Creeping)	80% - 89%	The scope is creeping. Backlog refinement or stakeholder feedback has introduced noticeable volume changes that require monitoring.
Red (Unstable)	0% - 79%	The scope is highly unstable. Significant changes threaten the predictability of the timeline and require immediate re-baselining.

2.4.6 Collaboration and Satisfaction (CS)

Collaboration and Satisfaction (CS) indicator evaluates the internal health of the project team and external happiness of project stakeholders. This metric combines three sub-scores:

- Collaboration
- Satisfaction
- Communication

All feedback is captured verbally during the sprint review and sprint retrospective. The Collaboration and Communication data is collected from the project team during the sprint retrospective, when the team reflects on how they worked together; during this session, the Product Owner verbally asks the team to rate three statements for collaboration and two statements for communication. Satisfaction data is collected from stakeholders during the sprint review, allowing them to evaluate value delivery after completed work has been presented; during this event, the Product Owner verbally asks the stakeholders to rate two statements.

Each sub-score converts its raw 1-5 verbal ratings into a normalized 0-1 score.

The formula is defined as:

$$CS = (Collaboration * 0.40) + (Satisfaction * 0.40) + (Communication * 0.20)$$

The rationale behind this distribution is structurally balanced:

- **Collaboration (40% Weight):** Team members rate three statements focusing on internal communication, conflict resolution, and knowledge sharing using a scale of 1 (Strongly Disagree) to 5 (Strongly Agree). The total sum of all ratings is divided by the total number of responses to establish the average, which is then divided by 5 to create a normalized decimal score.
- **Satisfaction (40% Weight):** Stakeholders rate two statements regarding team engagement and value delivery on the same 1-5 scale. The total sum of ratings is divided by the total responses and normalized by dividing by 5.
- **Communication (20% Weight):** Team members rate two statements concerning ceremony effectiveness and update clarity on the 1-5 scale. Responses are averaged and normalized by dividing by 5.

Collaboration and Satisfaction are each weighted at 40% because they capture two distinct, essential perspectives of delivery health (team-internal vs. stakeholder-external). Communication is weighted at 20% because it acts as an enabling factor rather than a direct outcome.

Example: Consider a sprint with 5 team members and a representative group of stakeholders:

- **Collaboration:** The team members provide 3 ratings each, generating 15 total responses. If these ratings sum to 63: Average = $63/15 = 4.2$, then Score = $4.2/5 = 0.84$ (84%)
- **Satisfaction:** The stakeholders provide responses yielding an average rating of 3.8: Score = $3.8/5 = 0.76$ (76%)
- **Communication:** The team members provide responses yielding an average rating of 4.5: Score = $4.5/5 = 0.90$ (90%)

$$CS = (84\% * 40\%) + (76\% * 40\%) + (90\% * 20\%) = 82\% \text{ (Damaged)}$$

The table below shows the three CS zones and what each means:

Zone	Range	Meaning
Engaged	90% - 100%	Team collaborates well. Stakeholders are satisfied.
Damaged	80% - 89%	Visible misalignment. Friction exists within the team or stakeholder satisfaction has decreased, requiring attention.
Broken	0% - 79%	Severe cultural or relationship breakdown. Immediate leadership intervention is required to repair collaboration or stakeholder trust.

2.4.7 Project Score and Decision Levels

Project Score (PS) is the most important key indicator used to show overall project health and is the average of all four indicators. Every indicator score is added up and divided by 4. The result (0% to 100%) maps to four decision levels, allowing Product Owner (PO) to determine whether to continue the project, escalate issues, or terminate the project.

The formula is defined as:

$$PS = (CDP + RE + SS + CS) / 4$$

The table below outlines the four thresholds, their corresponding levels, and the actions required:

Level	PS Range	Action
Proceed	90% - 100%	All indicators are Green. Project is on track.
Watch	80% - 89%	One or more indicators have shifted to Amber. Monitor closely.
Review	70% - 79%	Significant issues. Corrective action needed.
Escalate	0% - 69%	Project viability is in doubt. Escalate to governance.

Example: Consider a project with four individual indicators that yield the following validated scores: **PS** = $(70\% + 85\% + 90\% + 80\%)/4 = 0.81$ (81%)

A final Project Score of 81% places the project into the Watch level. This status alerts the Product Owner that while the project may proceed, specific underlying constraints require close monitoring to prevent a further drop into the Review zone.

2.4.8 Forecasting Engine

The forecasting engine utilizes current project metrics to predict the future. The Product Owner executes this calculation as part of the project health check, subsequently presenting the results in the next sprint planning session.

Forecast Completion Point (FCP)

The forecast Completion Point (FCP) estimates the total number of sprints a project will require based on the gap between the cumulative planned progress and the cumulative actual progress, as calculated by the CDP indicator.

The formula is defined as:

$$\text{FCP} = \text{Total Number of Sprints} + ((\text{PP_cum} - \text{AP_cum}) * \text{Total Number of Sprints})$$

Example:

Total Number of Sprints = 10

Current Status: The project is at Sprint 5 of 10, so PP_cum = 50%.

AP_cum at sprint 5 = 30%.

$$\text{FCP} = 10 + ((50\% - 30\%) * 10) = 12 \text{ sprints.}$$

The project is forecasted at a total of 12 sprints to reach completion.

2.4.9 MoSCoW Prioritization with APDS Scoring

Requirements (Work Units) are prioritized using the MoSCoW technique: Must Have, Should Have, Could Have, and Won't Have. The APDS framework makes this technique measurable by assigning a numerical value score to every requirement, allowing the project team to quantify the priority distribution and track how it shifts over time.

Each Work Unit is analyzed and assigned a value score ranging from 1 to 100 based on business value, strategic alignment, stakeholder impact, and other critical parameters. The recommended score ranges are:

- Must Have (80-100)
- Should Have (50-79)
- Could Have (20-49)
- Will Not Have: Items are not scored because they are out of scope

The APDS framework utilizes three MoSCoW-specific metrics to evaluate the project requirements:

Metric	Formula	What it does mean
MoSCoW Distribution Ratio (MDR)	Must Have count / Total requirements	The fraction of the backlog that represents committed, non-negotiable scope.
MoSCoW Value Density (MVD)	Total Value Score / Total requirements	The average of business value per requirement in the project backlog.
MoSCoW Stability Index (MSI)	$1 - (\text{Requirements with Changed category} / \text{Total Requirements})$	Priority stability over time. A score of 1 (100%) means no priorities changed.

Example: A project has 40 Work Units. Each category uses the average score of its range:

20 Must Have (average 90), 12 Should Have (average 65), 8 Could Have (average 35)

- **Total Value Score** = $(20 * 90) + (12 * 65) + (8 * 35) = 2,860$
- **MDR** = $20/40 = 0.50$ (50% of Work Units are in Must Have category)
- **MVD** = $2,860/40 = 71.5$ (average Work Unit value score is 71.5)
- **MSI** = If 4 Work Units reprioritized into a different MoSCoW category during execution, the metric will be: $1 - (4/40) = 0.90$ (90% stable)

MoSCoW Zone Classification

MDR and MVD are interpreted completely independently. MDR evaluates commitment balance, while MVD evaluates the concentration of business value. The underlying principle is that a healthy project needs both a committed core and a flexible buffer to survive changes.

MDR Zone Breakdown

The MDR range directly determines how much scope expansion of the release plan can absorb before the timeline breaks.

Zone	MDR Range	Meaning & Impact on the release Plan
Balanced	30% - 60%	Healthy distribution. Leaves adequate room to absorb changes. The team can defer lower-priority Work Units when new requests arrive, keeping delivery dates stable without adding extra sprints.
Top-Heavy	> 60%	Rigid backlog. Too many Must Haves destroy the safety buffer. New requirements force the team to either manually swap out existing critical items or pile Work Unit onto an overloaded team. This delays delivery.
Thin	< 30%	Weak core. Too few Must Haves signal a lack of clear direction. The committed core is too small to define a reliable minimum viable delivery, forcing the team into constant rework and mid-project scope redefinition as they struggle to uncover true stakeholder needs.

MVD zone Breakdown

The MVD score directly impacts whether the project team is spending its sprint capacity on work that genuinely delivers business value.

Zone	MVD Range	Meaning & Impact on the release Plan
Healthy	>= 50%	High-value backlog. The backlog sits in Should Have or Must Have territory. This proves the team is actively prioritizing for value, ensuring sprint capacity is spent on impactful deliverables.
Low-Value	< 50%	Low-value backlog. The backlog slips into Could Have territory, meaning low-value work dominates. The team spends capacity on items with marginal business value while critical Work Unit sits undelivered. This delays project benefits and extends duration by pushing critical scope into later, highly compressed sprints.

Note on Calibration: Zone thresholds are recommended defaults. Teams should calibrate these benchmarks against their own historical data after 3 to 4 sprints. While breakpoints may shift depending on project context, the core principle remains:

- MDR manages timeline risk against scope drift.
- MVD ensures the backlog maintains adequate business value.

MDR & Scope Stability (SS)

When MDR exceeds 60%, the product backlog becomes top-heavy. For instance, a backlog with 70% Must Haves has almost no flexible scope. When stakeholders request something new, there are not enough Should/Could Have requests to defer. The team should instead reprioritize either swapping one critical item for another or piling the new work onto an already full sprint. Each cycle of swaps or additions widens the gap between original and current scope, which drops Scope Stability (SS) into Creeping (80-89%) or Unstable (below 80%), ultimately destabilizing the release plan.

MSI & Collaboration and Satisfaction (CS)

The MSI tracks priority alignment. If requirements frequently shift between MoSCoW categories, it signals a fundamental disagreement among stakeholders regarding what matters most. This constant re-prioritization forces team members to waste effort refining items that eventually get deprioritized. The resulting friction undermines trust and lowers stakeholder satisfaction. Consequently, Collaboration and Satisfaction (CS) scores decline across communication, collaboration, and satisfaction sub-scores, driving the project toward a Damaged (80%-89%) or Broken (below 80%) state.

MVD & Financial Health

A low MVD (below 50) alerts management that the backlog has been dominated with low-value, minor Work Units. If this drop in value density occurs, it signals a value-for-money problem that should trigger a comprehensive scope re-evaluation before additional budget is authorized.

3. The APDS Lifecycle: Setup, Refinement, and Planning

The APDS framework does not start with the first sprint; it begins the moment a project is defined. This section covers the foundational work required before the first sprint to establish the baseline data that APDS needs for the entire project life cycle. Establishing this data early provides the team with a clear line of sight and complete, end-to-end traceability from the initial charter all the way to final delivery.

The Product Owner (PO) begins by creating a Project Charter, Product Vision, or Business Requirements Document (BRD) to capture high-level project information. From there, detailed requirements emerge through interviews, workshops, and document analysis.

To move from this initial kickoff to an active sprint backlog, the framework guides the team through the following steps:

- **Log all requirements:** Gather all requirements as candidate Work Units.
- **Prioritize the requirements:** Use MoSCoW analysis to rank these Work Units based on business value, risks, and dependencies.
- **Map out the release plan:** Organize the prioritized Work Units into a structured release plan that details the planned delivery order across sprints.
- **Build the product backlog:** Create the formal product backlog directly from the project release plan. Apply initial Fibonacci estimation to Work Units to determine their initial estimated durations, then identify and map risks and dependencies in the RAID log to assist with project prioritization.
- **Kick off Sprint Planning:** Gather the team to pull the top-priority items from the backlog into the first sprint.

Release Planning

Once Work units are established and prioritized using the MoSCoW technique, the next step involves mapping them into a release plan. The release plan acts as an interval distributed timeline, determining exactly when the prioritized Work Units will be delivered. This plan directly establishes the overall sprint count and baseline budget, and any identified dependencies are formally recorded in the RAID log. Once the release plan is developed, the product backlog is formalized and prioritized to set the stage for sprint planning, where these mapped requirements are committed to active sprints.

Product Backlog

The product backlog serves as the foundational data source for the APDS framework. Within the scrum framework, it is an ordered list of everything the project must deliver, prioritized based on MoSCoW APDS scoring. APDS does not replace this artifact; instead, it establishes an analytical tracking layer around it, turning every backlog item into a structured data point.

What APDS Requires from the Product Backlog

Every backlog item is classified as a Work Unit (WU). Each Work Unit requires the following three attributes:

- **Fibonacci Duration Estimate:** The project team collectively assign a Fibonacci number (1, 2, 3, 5, 8, 13, ...) to represent the estimated days of effort required. For example, a score of 1 represents 1 day of effort, while a 2 represents 2 days. Once established, this estimated duration remains fixed unless scope change occurs.
- **An Initial Rung Assignment:** Every new Work Unit begins at R1 (Draft) and progresses sequentially through R2 (Assembly), R3 (Validation), and R4 (Certified), until it reaches R5 (Agreed/Done).
- **Recorded Dependencies:** All internal or external dependencies including relationships to other Work Units, cross-functional teams, or external vendors should be identified and recorded in the RAID log from day one.

At this stage, no complex calculations are required. The focus is strictly on establishing the clean baseline data that downstream APDS processes will utilize.

Product Backlog Refinement

While the product backlog is established at the start of a project, it is a dynamic artifact rather than a static document. It is designed to evolve continuously throughout the project lifecycle as new information, feedback, and Work Units emerge. To ensure the backlog remains accurate and actionable, backlog refinement sessions take place during each sprint.

This iterative process ensures that upcoming Work Units are progressively elaborated, prioritized, and aligned with core project goals before entering sprint planning. During refinement, the team reviews and reorders items to ensure high-value work remains at the top. APDS supports this process by providing objective metrics to guide priority ordering, replacing intuition with empirical evidence.

The guiding principle is straightforward: the APDS indicators (CDP, RE, SS, and CS) are calculated based on the previous sprint and directly guide upcoming prioritization using MoSCoW APDS scoring.

How Indicators Guide Priority

The table below maps indicator health zones to clear priority actions. This framework serves as an objective guide to focus team discussions during backlog refinement:

Priority	Condition	Action
1 (High)	Any indicator in red zone (SS=Unstable, CS=Broken, RC=Critical, DP=Critical, RE=Critical)	Address immediately. Backlog priority should pivot to resolve this critical project risk.
2 (Medium)	Any indicator in amber zone (SS=Creeping, CS=Damaged, RC=Emerging, DP=Behind, RE=Over-Budget)	Investigate. Backlog ordering should incorporate targeted corrective actions.
3 (Low)	All indicators in green zone	No urgent action required. Maintain standard value-driven prioritization.

Sprint Planning

Sprint planning is the event where the team determines the upcoming sprint goal and defines the plan to achieve it. The APDS framework introduces two key elements to this collaborative event:

- Planned Progress (PP)
- Sprint Budget (SB)

The principle is straightforward: before the sprint begins, the Product Owner determines the Planned Progress as well as the Sprint Budget. The Product Owner presents the four indicators, the project score, and the Forecast Completion Point (FCP) from the previous sprint, along with a recovery plan from problem solving workshop if applicable. Based on these inputs, the team utilizes MoSCoW prioritization and collaborates with the Product Owner to prioritize the release plan, which directly feeds the product backlog. The team then selects Work Units from this product backlog and places them in their sprint backlog.

Performance indicators from the previous sprint and the project score directly inform what is committed to the upcoming sprint. If the Cumulative Delivery Pace (CDP) was Critical, a lighter load should be considered. If the Resource Efficiency (RE) was Critical, the team should investigate resourcing issues. If Scope Stability (SS) is Creeping or Unstable, scope growth should be factored into the upcoming sprint commitment, meaning the team may need to pull fewer Work Units to match the expanding scope. If Collaboration and Satisfaction (CS) is Damaged or Broken, capacity planning alone will not fix the problem; collaboration or satisfaction issues should be addressed before the team can work at full efficiency. Furthermore, if a recovery plan is generated from a problem-solving workshop, its specific action items and structural adjustments should directly feed into the spring planning ceremony.

4. Problem Solving Workshop

When the Project Score (PS) falls into the Watch, Review, or Escalate thresholds, Product Owner should pause routine progression and resolve underlying issues rather than pushing forward blindly. A dedicated problem-solving workshop is convened to identify root causes, establish consensus on corrective actions, and define an explicit timeline for improvement.

PS Level	PS Range	Workshop Type
Proceed	90% - 100%	No workshop is needed. Standard delivery and routine monitoring continue.
Watch	80% - 89%	A meeting is held with the core delivery team only. The session focuses exclusively on the affected indicator(s) to isolate root causes and assign quick-win corrective actions.
Review	70% - 79%	A comprehensive workshop is convened, bringing together both the delivery team and external stakeholders. Participants review all performance indicators to build a formal recovery plan with clear item owners.
Escalate	0% -69%	The project status is immediately escalated to executive governance or steering committees. A specialized workshop session is conducted to generate data that directly informs the final continuation or termination decision.

The problem-solving workshop is treated as an isolated event with a distinct purpose, rather than being combined with or absorbed by the standard sprint review or sprint retrospective ceremonies. The standard progression of events flows as follows:

- **Sprint Retrospective:** The delivery team meets internally to discuss team collaboration, interpersonal processes, and continuous improvements.
- **Problem-Solving Workshop:** The designated participants dig deeply into the structural root causes of the underperforming metrics and draft a formal recovery plan.
- **Sprint Planning:** The output of the recovery plan is fed directly into the upcoming sprint planning session, ensuring that corrective tasks are prioritized and capacity is allocated for remediation.

The deliverable of the problem-solving workshop is a structured recovery plan. This document should define:

- **Specific Corrective Actions:** The concrete tasks required to remediate the underperforming metrics.
- **Review Date:** A firm deadline for evaluating the effectiveness of the interventions.

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